TAISE Sample Questions

Test Yourself Before Taking the Exam

*The answer key, including justifications for each answer, begins on page 18

Module 1: Introduction to AI

Question M1-Q1

What is the historical significance of the 1956 Dartmouth Conference?

A. It standardized the transformer architecture for natural language processing tasks.
B. It debuted the first commercially available neural network hardware for pattern recognition.
C. It is recognized as the founding event that launched Artificial Intelligence as a field.
D. It introduced the first diffusion model for automated image generation.

Question M1-Q2

A team trains a model to label emails as "malware" or "not malware." What kind of model is this?

A. A discriminative model that learns to distinguish between classes in labeled data.
B. A diffusion model that reverses a noise process to reconstruct original inputs.
C. A generative model that creates new synthetic emails for augmented training.
D. A reinforcement model that optimizes labeling actions through environmental reward signals.

Question M1-Q3

An organization wants to use AI tools without managing infrastructure or runtime. Which cloud model fits best?

A. Platform as a Service (PaaS) requiring application runtime and middleware management.
B. Software as a Service (SaaS) that provides ready-to-use applications.
C. Infrastructure as a Service (IaaS) with full operating system and network control.
D. On-premises deployment managed and maintained entirely by the organization.

Question M1-Q4

What is the main role of the attention mechanism in transformers?

A. It removes the need for tokenization by directly processing raw text inputs.
B. It generates images from pure noise through iterative denoising steps.
C. It replaces GPUs by compressing model weights into smaller representations.
D. It learns which input elements are most important within the context window.

Question M1-Q5

A startup wants maximum flexibility to customize OS and applications for training an LLM, but does not want to buy servers. Which option is the most appropriate trade-off?

A. Use PaaS to avoid runtime responsibilities while controlling hypervisors.
B. Use SaaS to avoid managing any part of the stack while keeping full control.
C. Use on-premises clusters to eliminate cloud costs entirely.
D. Use IaaS to rent compute while retaining control over OS and application layers.

Module 2: Generative AI Architecture and Design

Question M2-Q1

Which of the following correctly distinguishes model architecture from services architecture?

A. Model architecture is only used for supervised learning; services architecture applies only to unsupervised learning.
B. Model architecture deals with internal design of AI models; services architecture focuses on external delivery systems.
C. Model architecture governs API routing and request handling; services architecture governs neural network layers.
D. Model architecture is maintained by DevOps teams; services architecture by ML researchers.

Question M2-Q2

Which model architecture introduced in 2017 replaced many RNN and CNN use cases in NLP?

A. GAN.
B. Transformer.
C. Diffusion model.
D. VAE.

Question M2-Q3

Which of the following best describes diffusion models?

A. They compress data into latent vectors and decode it through probabilistic sampling.
B. They progressively add noise and then learn to reverse it to generate new data.
C. They rely on adversarial training between a generator and a discriminator network.
D. They predict the next token in a sequence using left to right autoregression.

Question M2-Q4

In the attention mechanism, which component represents the information being advertised or offered?

A. Query.
B. Key.
C. Value.
D. Embedding.

Question M2-Q5

Which architectural feature makes transformers faster to train than RNNs?

A. Larger weight matrices that enable deeper layer representations.
B. Reliance on convolutional filters applied across token sequences.
C. Direct parallel token-to-token attention instead of step-by-step updates.
D. Sequential token processing that propagates information through hidden states.

Module 3: AI Use Cases: GenAI, Multimodal, and AI Agents

Question M3-Q1

Deepfakes are best defined as:

A. Compressed video files optimized for efficient cloud storage and streaming.
B. Synthetic media created using AI to mimic real people in audio or video.
C. Preprocessing methods for multimodal feature extraction and alignment.
D. Standard bias mitigation techniques applied during model fairness evaluations.

Question M3-Q2

What distinguishes AI safety from AI security in the module's framing?

A. Safety focuses on legal policy and regulatory compliance; security focuses on research ethics and peer review.
B. Safety handles access control and authentication; security covers hallucinations and output quality.
C. Safety addresses unintended harms and behavior; security defends against deliberate attacks or misuse.
D. Safety is about encryption and data protection; security is about empathy and user experience in chatbots.

Question M3-Q3

Which of the following is NOT a bias category emphasized in the module?

A. Deployment bias.
B. Computational bias.
C. Algorithmic bias.
D. Sampling bias.

Question M3-Q4

In healthcare, what is a key benefit of multimodal AI?

A. Correlating diverse inputs like imaging, genomics, and clinical notes for better predictions.
B. Limiting data sources to a single modality in order to simplify model explainability.
C. Replacing physicians with fully unsupervised model decisions across clinical settings.
D. Using only patient self-reported data and surveys as input to reduce sampling bias.

Question M3-Q5

What is the main purpose of a responsible AI life cycle?

A. To ensure ethics, transparency, fairness, and continuous monitoring across all stages.
B. To reduce GPU costs and compute overhead during model training phases.
C. To speed up model training and deployment timelines regardless of associated risks.
D. To limit AI deployment exclusively to healthcare and clinical use cases.

Module 4: Fairness, Accountability, and Explainability in AI

Question M4-Q1

What is the main purpose of explainability techniques like SHAP and LIME?

A. To provide understandable justifications for model predictions.
B. To improve computational efficiency of models.
C. To create inherently interpretable models.
D. To reduce training data requirements.

Question M4-Q2

Why was the skin lesion classifier case significant in the context of explainability?

A. It proved that only larger datasets can remove bias.
B. It demonstrated that neural networks always outperform interpretable models.
C. It revealed the model was detecting rulers instead of medical features.
D. It showed explainability tools mislabel cancer cases.

Question M4-Q2

Which fairness type ensures similar individuals are treated in similar ways by the model?

A. Group fairness.
B. Statistical parity.
C. Equalized odds.
D. Individual fairness.

Question M4-Q4

Which stakeholder group is most likely to need plain-language explanations and visual aids rather than technical details?

A. Regulators and auditors.
B. Technical security reviewers.
C. Non-technical business leaders.
D. Data scientists and ML engineers.

Question M4-Q5

What is the key benefit of model cards for AI transparency?

A. They replace the need for stakeholder engagement and external auditing processes.
B. They provide structured documentation of intended use, limitations, and performance.
C. They guarantee fairness outcomes across all demographic groups and use cases.
D. They reduce the computational cost of training and deploying production models.

Module 5: AI Model Lifecycle and Threat Taxonomy

Question M5-Q1

An attacker inserts a hidden trigger phrase into many training samples so the model misclassifies any input containing that phrase. What attack is this?

A. Logic bomb triggered by user identity.
B. Membership inference at evaluation time.
C. Backdoor attack during training.
D. Evasion attack against the deployed model.

Question M5-Q2

Which statement best defines concept drift?

A. Model weights are deliberately altered by an attacker.
B. The training set is replaced with adversarial data.
C. Two features are perfectly correlated in the dataset.
D. The relationship between inputs and outputs changes over time.

Question M5-Q3

Which NIST AI RMF function establishes organizational accountability, policies, and oversight for AI risk?

A. MANAGE.
B. GOVERN.
C. MAP.
D. MEASURE.

Question M5-Q4

Which attack attempts to tell whether a specific record was included in the model's training set?

A. Model manipulation.
B. Model extraction.
C. Membership inference.
D. Data poisoning.

Question M5-Q5

Which standard specifically provides guidance for AI risk management?

A. ISO/IEC 5338:2023.
B. ISO/IEC 23894:2023.
C. ISO/IEC 42001:2023.
D. ISO/IEC 27001:2022.

Module 6: Model Governance, Risk Management, and Compliance

Question M6-Q1

The NIST AI Risk Management Framework is organized around which four core functions?

A. PLAN, BUILD, TEST, DEPLOY.
B. COLLECT, ANALYZE, REPORT, VERIFY.
C. IDENTIFY, PROTECT, DETECT, RESPOND.
D. GOVERN, MAP, MEASURE, MANAGE.

Question M6-Q2

Under the EU AI Act's phased enforcement, which category is prohibited immediately (Phase One)?

A. Every AI application deployed in public sector organizations and government agencies.
B. All general-purpose foundation models regardless of their assessed risk level or application.
C. Only systems involving any biometric data collection, without exception.
D. Systems using subliminal techniques or biometric categorization of sensitive attributes.

Question M6-Q3

What is the main purpose of a RACI matrix in AI governance programs?

A. Classify AI systems into prohibited, high-risk, or minimal-risk categories per regulatory requirements.
B. Score AI models for fairness and bias using quantitative metrics and statistical tests.
C. Define security controls at each infrastructure layer for cloud-hosted AI services.
D. Assign Responsible, Accountable, Consulted, and Informed roles for key governance activities.

Question M6-Q4

Which U.S. jurisdiction has Automated Decision-Making Technology (ADMT) rules attached to its privacy law?

A. California, via CCPA ADMT regulations requiring notices, rights, and risk assessments.
B. Texas, under a statewide AI licensing board that oversees all models.
C. Washington, with a blanket ban on automated decisions across industries.
D. New York, through a federal mandate that preempts state privacy statutes.

Question M6-Q5

Which trustworthiness characteristic is emphasized by the NIST AI RMF?

A. Lowest possible cost regardless of impact.
B. Unlimited model complexity to improve accuracy.
C. Valid and reliable performance under expected conditions.
D. Fastest deployment speed without testing.

Module 7: Introduction to AI Safety and Security

Question M7-Q1

What is the primary distinction between AI safety and AI security?

A. Safety focuses on cost reduction; security focuses on compliance only.
B. Safety prevents unintended harm; security protects against intentional attacks.
C. Safety applies only to physical systems; security applies only to digital systems.
D. Safety is regulated in all cases; security is voluntary in most jurisdictions.

Question M7-Q2

In the enterprise architecture framework, monitoring for hallucinations and documenting low-confidence outputs primarily belongs to which layer?

A. Application/Model layer.
B. Governance layer.
C. Data and Information layer.
D. Operation layer.

Question M7-Q3

Which option best defines AI alignment?

A. Ensuring AI models remain within budget constraints and resource allocation targets.
B. Automating safety checks and validation processes without requiring human oversight.
C. Mapping AI outputs to specific enterprise architecture layers and integration points.
D. Keeping AI goals and behavior consistent with human values and intended outcomes.

Question M7-Q4

Which pillar of the Four Pillars of AI Security addresses data poisoning and privacy leakage risks?

A. Vulnerability Management.
B. Data Security & Privacy Protection.
C. Model Security.
D. Governance & Compliance.

Question M7-Q5

Which scenario best illustrates “AI for security” (as opposed to “security for AI”)?

A. Implementing Zero Trust access controls and authentication for model serving APIs.
B. Using AI to detect anomalies in network traffic logs across the enterprise.
C. Running prompt injection detection and input sanitization on LLM request inputs.
D. Encrypting data at rest and in transit within AI training and inference pipelines.

Module 8: Cloud and AI Security

Question M8-Q1

In the shared responsibility model, who is primarily responsible for securing data at rest in an IaaS environment?

A. Customer only.
B. Both provider and customer equally.
C. Third-party auditors.
D. Cloud service provider only.

Question M8-Q2

Why is automatic key rotation in a Key Management Service (KMS) important for AI workloads?

A. It reduces model latency during inference by offloading cryptographic operations.
B. It ensures encryption keys are regularly updated to limit exposure risk.
C. It eliminates the need for access policies by automating credential management.
D. It simplifies API integration with AI pipelines by standardizing key formats.

Question M8-Q3

During an AI model breach, logs show repeated abnormal queries with slight input variations. Which type of attack does this indicate?

A. Policy misconfiguration.
B. Hardware failure.
C. Evasion attack.
D. Data poisoning.

Question M8-Q4

In a secure MLOps pipeline, what is the primary role of dependency pinning and lockfiles?

A. Accelerating GPU utilization across distributed training workloads.
B. Improving CI/CD performance metrics and deployment speed.
C. Reducing infrastructure costs by consolidating package repositories.
D. Preventing unauthorized or unexpected third-party code updates.

Question M8-Q5

You are designing access policies for an AI inference service. Which strategy BEST reflects Zero Trust principles?

A. Apply dynamic, per-session access policies that evaluate identity, context, and risk before granting access.
B. Reduce monitoring and logging overhead to minimize operational costs.
C. Provide static API keys with no expiration for simplified integration.
D. Grant persistent access to all authenticated users within the network.

Module 9: Data Security & Privacy in AI Systems

Question M9-Q1

Which of the following best describes data lineage in AI systems?

A. Tracking the origin, transformations, and usage of data throughout its lifecycle.
B. Creating synthetic datasets to mimic the statistical properties of real data.
C. Restricting access to data through role-based access control policies.
D. Encrypting sensitive data fields before storage in production databases.

Question M9-Q2

Why is anonymization critical in AI systems that handle healthcare data?

A. It guarantees data completeness across all fields and records in the dataset.
B. It eliminates the need for encryption by removing identifiable information entirely.
C. It improves model training speed by reducing the volume of input features.
D. It protects individual identities and ensures compliance with laws like HIPAA.

Question M9-Q3

What distinguishes a backdoor attack from other poisoning attacks?

A. It embeds a hidden trigger that causes targeted misclassification.
B. It only occurs in image datasets and computer vision models.
C. It always requires direct access to model metadata and configuration.
D. It increases training speed by reducing the effective dataset size.

Question M9-Q4

What is one advantage of using synthetic data in training AI systems?

A. It guarantees zero bias in the dataset.
B. It reduces privacy risks by avoiding direct use of sensitive records.
C. It improves query response speed.
D. It eliminates the need for governance policies.

Question M9-Q5

During an audit, a regulator requests proof that sensitive features were flagged and removed before model training. Which governance control provides this evidence?

A. Secure API logging of inference requests.
B. Imputation of missing values during processing.
C. PCA dimensionality reduction before feature selection.
D. Metadata management with sensitive field tagging.

Module 10: Continuous Learning and Adaptation

Question M10-Q1

What is the main purpose of continuous learning in AI systems?

A. To ensure models adapt to new data while retaining prior knowledge.
B. To maximize training speed and throughput regardless of accuracy.
C. To reduce model size and parameter count for faster inference at deployment.
D. To eliminate the need for ongoing monitoring, retraining, or manual updates.

Question M10-Q2

Which scenario best illustrates prediction drift?

A. The training dataset becomes corrupted.
B. The outputs of a model shift significantly compared to historical predictions.
C. Input features of a fraud detection system change over time.
D. New features are added to a dataset mid-deployment.

Question M10-Q3

What is catastrophic forgetting in continuous learning?

A. When a model loses prior knowledge after retraining on new data.
B. When a system's database becomes corrupted due to storage/hardware failures.
C. When bias testing is skipped during the model validation and evaluation.
D. When model weights cannot be updated due to hardware limitations.

Question M10-Q4

Why would a healthcare consortium adopt federated learning instead of centralizing data?

A. To eliminate the need for monitoring and compliance oversight across institutions.
B. To increase training speed by pooling patient data into a single central repository.
C. To reduce compute costs by distributing workloads across organizations.
D. To protect patient privacy while still enabling collective model training.

Question M10-Q5

Which feedback mechanism ensures expert input is weighted more heavily than casual user input?

A. Contextual feedback analysis.
B. Random sampling of feedback.
C. Batch updates every quarter.
D. Consensus by majority voting.

Answer Key & Justifications

Module 1: Introduction to AI

Question M1-Q1

What is the historical significance of the 1956 Dartmouth Conference?

A. It standardized the transformer architecture for natural language processing tasks.
B. It debuted the first commercially available neural network hardware for pattern recognition.
C. It is recognized as the founding event that launched Artificial Intelligence as a field. ✓
D. It introduced the first diffusion model for automated image generation.

Justifications:

A. Transformers were introduced in 2017, not in 1956.
B. No commercial neural network hardware was launched at that conference.
C. Module history notes the Dartmouth conference as the point where AI emerged as a formal research field.
D. Diffusion models are modern; they were not introduced in 1956.

Question M1-Q2

A team trains a model to label emails as "malware" or "not malware." What kind of model is this?

A. A discriminative model that learns to distinguish between classes in labeled data. ✓
B. A diffusion model that reverses a noise process to reconstruct original inputs.
C. A generative model that creates new synthetic emails for augmented training.
D. A reinforcement model that optimizes labeling actions through environmental reward signals.

Justifications:

A. The task uses labeled data to separate classes, which matches the definition of discriminative (predictive) ML.
B. Diffusion models target image/video synthesis, not email classification per se.
C. Generative models create new data; here the goal is classification, not generation.
D. Reinforcement learning optimizes actions via rewards; this scenario is supervised classification.

Question M1-Q3

An organization wants to use AI tools without managing infrastructure or runtime. Which cloud model fits best?

A. Platform as a Service (PaaS) requiring application runtime and middleware management.
B. Software as a Service (SaaS) that provides ready-to-use applications. ✓
C. Infrastructure as a Service (IaaS) with full operating system and network control.
D. On-premises deployment managed and maintained entirely by the organization.

Justifications:

A. PaaS reduces but does not eliminate platform responsibilities.
B. SaaS delivers complete applications with minimal customer management burden.
C. IaaS requires managing OS and application layers.
D. On-premises maximizes management effort rather than minimizing it.

Question M1-Q4

What is the main role of the attention mechanism in transformers?

A. It removes the need for tokenization by directly processing raw text inputs.
B. It generates images from pure noise through iterative denoising steps.
C. It replaces GPUs by compressing model weights into smaller representations.
D. It learns which input elements are most important within the context window. ✓

Justifications:

A. Transformers still operate over tokens; attention does not replace them.
B. Image generation from noise describes diffusion, not attention itself.
C. Attention does not replace hardware; GPUs enable parallel training.
D. Attention weighs relevance of tokens to improve representation and generation.

Question M1-Q5

A startup wants maximum flexibility to customize OS and applications for training an LLM, but does not want to buy servers. Which option is the most appropriate trade-off?

A. Use PaaS to avoid runtime responsibilities while controlling hypervisors.
B. Use SaaS to avoid managing any part of the stack while keeping full control.
C. Use on-premises clusters to eliminate cloud costs entirely.
D. Use IaaS to rent compute while retaining control over OS and application layers. ✓

Justifications:

A. PaaS abstracts runtime; control over low-level components is limited.
B. SaaS minimizes control and customization; it does not grant full control.
C. On-premises contradicts the constraint of not purchasing servers.
D. IaaS offers virtualized infrastructure with customer control of OS/apps matching the stated need.

Module 2: Generative AI Architecture and Design

Question M2-Q1

Which of the following correctly distinguishes model architecture from services architecture?

A. Model architecture is only used for supervised learning; services architecture applies only to unsupervised learning.
B. **Model architecture deals with internal design of AI models; services architecture focuses on external delivery systems. ✓**
C. Model architecture governs API routing and request handling; services architecture governs neural network layers.
D. Model architecture is maintained by DevOps teams; services architecture by ML researchers.

Justifications:

A. Both architectures apply broadly, not tied to supervised/unsupervised only.
B. Model architecture = internal neural design; services = external delivery, APIs, scaling.
C. This inverts responsibilities; API routing is part of services.
D. Both DevOps and researchers can influence both; division is inaccurate.

Question M2-Q2

Which model architecture introduced in 2017 replaced many RNN and CNN use cases in NLP?

A. GAN.
B. Transformer. ✓
C. Diffusion model.
D. VAE.

Justifications:

A. GANs are adversarial generative architectures, not NLP-focused replacements.
B. Transformers revolutionized NLP with parallelism and self-attention.
C. Diffusion models are for generative image/video tasks.
D. VAEs compress and decode; they did not replace RNNs in NLP.

Question M2-Q3

Which of the following best describes diffusion models?

A. They compress data into latent vectors and decode it through probabilistic sampling.
B. They progressively add noise and then learn to reverse it to generate new data. ✓
C. They rely on adversarial training between a generator and a discriminator network.
D. They predict the next token in a sequence using left to right autoregression.

Justifications:

A. Latent vector probabilistic decoding matches VAEs.
B. Diffusion models learn a reverse denoising process for content generation.
C. Adversarial training describes GANs, not diffusion models.
D. Autoregressive models predict tokens sequentially.

Question M2-Q4

In the attention mechanism, which component represents the information being advertised or offered?

A. Query.
B. Key. ✓
C. Value.
D. Embedding.

Justifications:

A. Queries ask about relevance, they don't advertise info.
B. Keys advertise what a token contains; queries match against them.
C. Values are the actual content returned, not the advertised property.
D. Embeddings represent tokens numerically, not their attention role.

Question M2-Q5

Which architectural feature makes transformers faster to train than RNNs?

A. Larger weight matrices that enable deeper layer representations.
B. Reliance on convolutional filters applied across token sequences.
C. Direct parallel token-to-token attention instead of step-by-step updates. ✓
D. Sequential token processing that propagates information through hidden states.

Justifications:

A. Deeper layers increase compute but don't explain faster training.
B. Convolutions apply to CNNs, not transformer speed advantage.
C. Transformers apply self-attention in parallel unlike RNN sequential bottlenecks.
D. Sequential token processing describes RNNs, not transformers.

Module 3: AI Use Cases: GenAI, Multimodal, and AI Agents

Question M3-Q1

Deepfakes are best defined as:

A. Compressed video files optimized for efficient cloud storage and streaming.
B. Synthetic media created using AI to mimic real people in audio or video. ✓
C. Preprocessing methods for multimodal feature extraction and alignment.
D. Standard bias mitigation techniques applied during model fairness evaluations.

Justifications:

A. Compression is unrelated to synthetic media generation and deception.
B. Deepfakes are AI-generated or manipulated media that convincingly imitate real individuals.
C. Preprocessing for multimodal learning is different from fabricating media content.
D. Deepfakes are an ethical challenge, not a mitigation technique.

Question M3-Q2

What distinguishes AI safety from AI security in the module's framing?

A. Safety focuses on legal policy and regulatory compliance; security focuses on research ethics and peer review.
B. Safety handles access control and authentication; security covers hallucinations and output quality.
C. Safety addresses unintended harms and behavior; security defends against deliberate attacks or misuse. ✓
D. Safety is about encryption and data protection; security is about empathy and user experience in chatbots.

Justifications:

A. Both domains span technical and governance measures; the option is an oversimplification.
B. Hallucinations are a safety issue; access control is a security mechanism—this reverses roles.
C. Safety ensures systems behave as intended and avoid harm; security protects systems from adversaries and misuse.
D. Encryption is a security control; empathy is not the defining focus of security versus safety.

Question M3-Q3

Which of the following is NOT a bias category emphasized in the module?

A. Deployment bias.
B. Computational bias. ✓
C. Algorithmic bias.
D. Sampling bias.

Justifications:

A. Deployment bias is explicitly included and defined.
B. The module highlights sampling, deployment, algorithmic, historical, evaluation, and label bias—not 'computational bias' as a category.
C. Algorithmic bias is explicitly included and defined.
D. Sampling bias is explicitly included and defined.

Question M3-Q4

In healthcare, what is a key benefit of multimodal AI?

A. Correlating diverse inputs like imaging, genomics, and clinical notes for better predictions. ✓
B. Limiting data sources to a single modality in order to simplify model explainability.
C. Replacing physicians with fully unsupervised model decisions across clinical settings.
D. Using only patient self-reported data and surveys as input to reduce sampling bias.

Justifications:

A. Multimodal fusion across radiology, pathology, and EHRs enables richer, more accurate clinical insights.
B. Reducing modalities sacrifices context and can harm performance.
C. Systems augment clinicians; responsible use keeps humans in the loop.
D. Exclusive reliance on self-reports ignores other valuable modalities and may increase bias.

Question M3-Q5

What is the main purpose of a responsible AI life cycle?

A. To ensure ethics, transparency, fairness, and continuous monitoring across all stages. ✓
B. To reduce GPU costs and compute overhead during model training phases.
C. To speed up model training and deployment timelines regardless of associated risks.
D. To limit AI deployment exclusively to healthcare and clinical use cases.

Justifications:

A. Responsible AI spans from problem framing through monitoring to manage harms, governance, and accountability.
B. Cost control is not the primary aim; governance and safety are.
C. Rushing deployment without governance contradicts responsible AI principles.
D. Responsible AI applies across sectors, not just healthcare.

Module 4: Fairness, Accountability, and Explainability in AI

Question M4-Q1

What is the main purpose of explainability techniques like SHAP and LIME?

A. To provide understandable justifications for model predictions. ✓
B. To improve computational efficiency of models.
C. To create inherently interpretable models.
D. To reduce training data requirements.

Justifications:

A. SHAP and LIME are post-hoc methods that explain complex models' behavior to users, auditors, and regulators.
B. They may be computationally costly; efficiency is not their primary purpose.
C. They explain black-box models; they do not turn them into inherently interpretable ones.
D. They do not affect the amount of training data required.

Question M4-Q2

Why was the skin lesion classifier case significant in the context of explainability?

A. It proved that only larger datasets can remove bias.
B. It demonstrated that neural networks always outperform interpretable models.
C. It revealed the model was detecting rulers instead of medical features. ✓
D. It showed explainability tools mislabel cancer cases.

Justifications:

A. Dataset size alone doesn't address spurious correlations; explainability is needed to detect them.
B. The case does not claim universal superiority of neural networks.
C. Explainability exposed a spurious correlation: the model used rulers as a proxy for malignancy.
D. The issue was the model's shortcut, not the explainability tools.

Question M4-Q3

Which fairness type ensures similar individuals are treated in similar ways by the model?

A. Group fairness.
B. Statistical parity.
C. Equalized odds.
D. Individual fairness. ✓

Justifications:

A. Group fairness focuses on aggregate parity across demographics, not individuals.
B. Statistical parity is an outcome rate constraint across groups.
C. Equalized odds constrains TPR and FPR parity across groups.
D. Individual fairness aims for person-level equivalence for similar cases.

Question M4-Q4

Which stakeholder group is most likely to need plain-language explanations and visual aids rather than technical details?

A. Regulators and auditors.
B. Technical security reviewers.
C. Non-technical business leaders. ✓
D. Data scientists and ML engineers.

Justifications:

A. Regulators need structured evidence like model cards and audits, not only high-level visuals.
B. Security reviewers need detailed technical and threat information.
C. Non-technical stakeholders benefit from high-level summaries, visuals, FAQs, and practical limitations.
D. Technical teams require model internals, metrics, and error analysis.

Question M4-Q5

What is the key benefit of model cards for AI transparency?

A. They replace the need for stakeholder engagement and external auditing processes.
B. They provide structured documentation of intended use, limitations, and performance. ✓
C. They guarantee fairness outcomes across all demographic groups and use cases.
D. They reduce the computational cost of training and deploying production models.

Justifications:

A. They do not replace stakeholder communication—they complement it.
B. Model cards standardize key information to support auditing, governance, and communication.
C. Documentation cannot guarantee fairness; it enables assessment and improvement.
D. Model cards do not affect training compute.

Module 5: AI Model Lifecycle and Threat Taxonomy

Question M5-Q1

An attacker inserts a hidden trigger phrase into many training samples so the model misclassifies any input containing that phrase. What attack is this?

A. Logic bomb triggered by user identity.
B. Membership inference at evaluation time.
C. Backdoor attack during training. ✓
D. Evasion attack against the deployed model.

Justifications:

A. Logic bombs embed malicious logic but not necessarily via a learned trigger phrase in data.
B. Membership inference aims to detect if records were in training data, not to misclassify via triggers.
C. Backdoors cause specific inputs with a trigger to bypass normal behavior.
D. Evasion crafts inputs at inference; here the training set was altered.

Question M5-Q2

Which statement best defines concept drift?

A. Model weights are deliberately altered by an attacker.
B. The training set is replaced with adversarial data.
C. Two features are perfectly correlated in the dataset.
D. The relationship between inputs and outputs changes over time. ✓

Justifications:

A. Deliberate weight changes describe model manipulation, not drift.
B. Replacing training data is a poisoning scenario, not drift.
C. Perfect correlation refers to multicollinearity, not drift.
D. Concept drift is a shift in the underlying mapping, degrading performance.

Question M5-Q3

Which NIST AI RMF function establishes organizational accountability, policies, and oversight for AI risk?

A. MANAGE.
B. GOVERN. ✓
C. MAP.
D. MEASURE.

Justifications:

A. MANAGE implements responses and continuous improvement.
B. GOVERN creates culture, policies, roles, and accountability across the lifecycle.
C. MAP focuses on context, stakeholders, and impacts.
D. MEASURE is about testing and monitoring risks.

Question M5-Q4

Which attack attempts to tell whether a specific record was included in the model's training set?

A. Model manipulation.
B. Model extraction.
C. Membership inference. ✓
D. Data poisoning.

Justifications:

A. Model manipulation alters weights or logic rather than inferring membership.
B. Model extraction steals functionality via queries but not membership status.
C. Membership inference probes model outputs to infer training membership.
D. Poisoning corrupts training data; it is not an inference attack.

Question M5-Q5

Which standard specifically provides guidance for AI risk management?

A. ISO/IEC 5338:2023.
B. ISO/IEC 23894:2023. ✓
C. ISO/IEC 42001:2023.
D. ISO/IEC 27001:2022.

Justifications:

A. 5338 covers AI engineering lifecycle practices rather than risk guidance focus.
B. 23894 gives AI-specific risk management guidance across the lifecycle.
C. 42001 is a certifiable AI management system standard, not risk guidance per se.
D. 27001 addresses information security management systems, not AI-specific risk.

Module 6: Model Governance, Risk Management, and Compliance

Question M6-Q1

The NIST AI Risk Management Framework is organized around which four core functions?

A. PLAN, BUILD, TEST, DEPLOY.
B. COLLECT, ANALYZE, REPORT, VERIFY.
C. IDENTIFY, PROTECT, DETECT, RESPOND.
D. GOVERN, MAP, MEASURE, MANAGE. ✓

Justifications:

A. These steps describe a generic delivery lifecycle, not the AI RMF.
B. These activities do not match the AI RMF structure.
C. Those are NIST Cybersecurity Framework functions, not the AI RMF core functions.
D. NIST AI RMF defines four functions that span culture, context, assessment, and risk response.

Question M6-Q2

Under the EU AI Act's phased enforcement, which category is prohibited immediately (Phase One)?

A. Every AI application deployed in public sector organizations and government agencies.
B. All general-purpose foundation models regardless of their assessed risk level or application.
C. Only systems involving any biometric data collection, without exception.
D. Systems using subliminal techniques or biometric categorization of sensitive attributes. ✓

Justifications:

A. Public sector deployments are not categorically banned; obligations depend on risk classification.
B. General-purpose models face obligations later; they are not universally banned in Phase One.
C. Biometric use is not blanket-banned; prohibitions target specific biometric categorization practices.
D. The Act bans unacceptable-risk systems such as subliminal manipulation and certain biometric categorization from the outset.

Question M6-Q3

What is the main purpose of a RACI matrix in AI governance programs?

A. Classify AI systems into prohibited, high-risk, or minimal-risk categories per regulatory requirements.
B. Score AI models for fairness and bias using quantitative metrics and statistical tests.
C. Define security controls at each infrastructure layer for cloud-hosted AI services.
D. Clarify accountability by assigning who is Responsible, Accountable, Consulted, and Informed for key activities. ✓

Justifications:

A. Risk classification is a regulatory activity; RACI is about roles and responsibilities.
B. Fairness scoring is a measurement activity, not a responsibility assignment tool.
C. Control definition is a security architecture task, not the purpose of RACI.
D. RACI establishes clear role assignments and decision rights to prevent accountability gaps.

Question M6-Q4

Which U.S. jurisdiction has Automated Decision-Making Technology (ADMT) rules attached to its privacy law?

A. **California, via CCPA ADMT regulations requiring notices, rights, and risk assessments. ✓**
B. Texas, under a statewide AI licensing board that oversees all models.
C. Washington, with a blanket ban on automated decisions across industries.
D. New York, through a federal mandate that preempts state privacy statutes.

Justifications:

A. California's CCPA ADMT rules add pre-use notices, consumer rights, and business requirements for significant decisions.
B. No such statewide AI licensing board exists in Texas within this module's scope.
C. Washington does not impose a blanket ban; requirements vary by context.
D. There is no federal preemption described that creates a New York-specific mandate here.

Question M6-Q5

Which trustworthiness characteristic is emphasized by the NIST AI RMF?

A. Lowest possible cost regardless of impact.
B. Unlimited model complexity to improve accuracy.
C. **Valid and reliable performance under expected conditions. ✓**
D. Fastest deployment speed without testing.

Justifications:

A. Cost is an operational consideration, not a trustworthiness characteristic in the AI RMF.
B. Complexity alone does not satisfy trustworthiness objectives and can harm explainability.
C. Trustworthiness includes validity, reliability, safety, security, resilience, accountability, transparency, and fairness.
D. Speed without testing undermines safety and reliability requirements.

Module 7: Introduction to AI Safety and Security

Question M7-Q1

What is the primary distinction between AI safety and AI security?

A. Safety focuses on cost reduction; security focuses on compliance only.
B. Safety prevents unintended harm; security protects against intentional attacks. ✓
C. Safety applies only to physical systems; security applies only to digital systems.
D. Safety is regulated in all cases; security is voluntary in most jurisdictions.

Justifications:

A. Cost and compliance are operational concerns, not the defining difference.
B. Safety focuses on avoiding unintended harm; security focuses on resisting deliberate attacks.
C. Both safety and security apply to physical and digital contexts; the distinction is not physical vs. digital.
D. Regulatory status varies; the key distinction is harm type (unintentional vs. intentional).

Question M7-Q2

In the enterprise architecture framework, monitoring for hallucinations and documenting low-confidence outputs primarily belongs to which layer?

A. Application/Model layer.
B. Governance layer.
C. Data and Information layer.
D. Operation layer. ✓

Justifications:

A. Application/Model covers robustness and error handling; this task is operational monitoring across use.
B. Governance sets policy and oversight rather than real-time monitoring.
C. Data and Information concerns data management and integrity, not outputs monitoring per se.
D. Operation focuses on day-to-day monitoring and safe user interactions, including tracking model outputs and confidence.

Question M7-Q3

Which option best defines AI alignment?

A. Ensuring AI models remain within budget constraints and resource allocation targets.
B. Automating safety checks and validation processes without requiring human oversight.
C. Mapping AI outputs to specific enterprise architecture layers and integration points.
D. Keeping AI goals and behavior consistent with human values and intended outcomes. ✓

Justifications:

A. Budget control is not the definition of alignment.
B. Automation can help, but alignment requires oversight and value consistency.
C. Architecture mapping is unrelated to value alignment.
D. Alignment addresses value-consistency and avoidance of reward hacking and unintended side effects.

Question M7-Q4

Which pillar of the Four Pillars of AI Security addresses data poisoning and privacy leakage risks?

A. Vulnerability Management.
B. Data Security & Privacy Protection. ✓
C. Model Security.
D. Governance & Compliance.

Justifications:

A. Vulnerability Management focuses on supply chain, infra vulns, and pipeline security.
B. The Data pillar covers training/inference data protection, minimizing leakage, and detecting poisoning.
C. Model Security focuses on model theft, inversion, adversarial examples, and backdoors.
D. Governance & Compliance addresses roles, policies, and oversight rather than data-specific risks.

Question M7-Q5

Which scenario best illustrates “AI for security” (as opposed to “security for AI”)?

A. Implementing Zero Trust access controls and authentication for model serving APIs.
B. Using AI to detect anomalies in network traffic logs across the enterprise. ✓
C. Running prompt injection detection and input sanitization on LLM request inputs.
D. Encrypting data at rest and in transit within AI training and inference pipelines.

Justifications:

A. Zero Trust for model APIs is security for AI (protecting the AI system).
B. AI for security applies AI techniques to defend external systems and infrastructure.
C. Prompt injection detection is a control for securing AI models, not using AI to protect other systems.
D. Data encryption in pipelines protects AI systems’ security for AI.

Module 8: Cloud and AI Security

Question M8-Q1

In the shared responsibility model, who is primarily responsible for securing data at rest in an IaaS environment?

A. Customer only. ✓
B. Both provider and customer equally.
C. Third-party auditors.
D. Cloud service provider only.

Justifications:

A. In IaaS, customers are responsible for securing data at rest using encryption and access controls.
B. The division of responsibilities is not equal; the customer handles in-cloud data.
C. Auditors evaluate compliance but are not responsible for security implementation.
D. Providers secure the infrastructure, not customer data in IaaS.

Question M8-Q2

Why is automatic key rotation in a Key Management Service (KMS) important for AI workloads?

A. It reduces model latency during inference by offloading cryptographic operations.
B. It ensures encryption keys are regularly updated to limit exposure risk. ✓
C. It eliminates the need for access policies by automating credential management.
D. It simplifies API integration with AI pipelines by standardizing key formats.

Justifications:

A. Rotation affects security, not model performance.
B. Regularly rotating keys reduces long-term risk and limits exposure if keys are compromised.
C. Access policies remain essential even with KMS.
D. API integration is unrelated to key rotation.

Question M8-Q3

During an AI model breach, logs show repeated abnormal queries with slight input variations. Which type of attack does this indicate?

A. Policy misconfiguration.
B. Hardware failure.
C. Evasion attack. ✓
D. Data poisoning.

Justifications:

A. Policy misconfiguration is not adversarial input.
B. Hardware failure is unrelated.
C. Evasion attacks use small input changes to bypass model detection.
D. Data poisoning occurs during training.

Question M8-Q4

In a secure MLOps pipeline, what is the primary role of dependency pinning and lockfiles?

A. Accelerating GPU utilization across distributed training workloads.
B. Improving CI/CD performance metrics and deployment speed.
C. Reducing infrastructure costs by consolidating package repositories.
D. Preventing unauthorized or unexpected third-party code updates. ✓

Justifications:

A. Not related to GPU utilization.
B. Not related to performance metrics.
C. Not for cost reduction.
D. Dependency pinning ensures consistent versions and reduces supply chain risks.

Question M8-Q5

You are designing access policies for an AI inference service. Which strategy BEST reflects Zero Trust principles?

A. Apply dynamic, per-session access policies that evaluate identity, context, and risk before granting access. ✓
B. Reduce monitoring and logging overhead to minimize operational costs.
C. Provide static API keys with no expiration for simplified integration.
D. Grant persistent access to all authenticated users within the network.

Justifications:

A. Zero Trust enforces dynamic, risk-based access.
B. Monitoring is essential.
C. Static keys increase risk.
D. Persistent access violates Zero Trust.

Module 9: Data Security & Privacy in AI Systems

Question M9-Q1

Which of the following best describes data lineage in AI systems?

- **A. Tracking the origin, transformations, and usage of data throughout its lifecycle. ✓**
- B. Creating synthetic datasets to mimic the statistical properties of real data.
- C. Restricting access to data through role-based access control policies.
- D. Encrypting sensitive data fields before storage in production databases.

Justifications:

- A. Data lineage is the process of documenting the origin, transformation, and use of data throughout its lifecycle.
- B. Synthetic data reduces privacy risks but does not record data history.
- C. RBAC manages permissions but is not equivalent to lineage tracking.
- D. Encryption protects confidentiality but does not track data usage history.

Question M9-Q2

Why is anonymization critical in AI systems that handle healthcare data?

- A. It guarantees data completeness across all fields and records in the dataset.
- B. It eliminates the need for encryption by removing identifiable information entirely.
- C. It improves model training speed by reducing the volume of input features.
- **D. It protects individual identities and ensures compliance with laws like HIPAA. ✓**

Justifications:

- A. Completeness is a data quality issue, not privacy.
- B. Encryption remains necessary even with anonymization.
- C. Anonymization is for privacy, not model performance.
- D. Anonymization prevents identification of individuals, ensuring compliance with privacy laws like HIPAA.

Question M9-Q3

What distinguishes a backdoor attack from other poisoning attacks?

A. **It embeds a hidden trigger that causes targeted misclassification. ✓**
B. It only occurs in image datasets and computer vision models.
C. It always requires direct access to model metadata and configuration.
D. It increases training speed by reducing the effective dataset size.

Justifications:

A. Backdoor attacks plant hidden triggers that cause specific outputs when activated.
B. Backdoor attacks can occur across modalities, not only in images.
C. They manipulate training data, not metadata.
D. They degrade security, not training performance speed.

Question M9-Q4

What is one advantage of using synthetic data in training AI systems?

A. It guarantees zero bias in the dataset.
B. **It reduces privacy risks by avoiding direct use of sensitive records. ✓**
C. It improves query response speed.
D. It eliminates the need for governance policies.

Justifications:

A. Synthetic data does not guarantee zero bias—it may perpetuate biases from source data or fail to capture rare or complex patterns.
B. Synthetic data reduces direct exposure of sensitive records, lowering re-identification risk while preserving data utility.
C. Faster query response is unrelated to the privacy or bias implications of synthetic data.
D. Governance policies are still required; synthetic data is not a substitute for oversight.

Question M9-Q5

During an audit, a regulator requests proof that sensitive features were flagged and removed before model training. Which governance control provides this evidence?

A. Secure API logging of inference requests.
B. Imputation of missing values during preprocessing.
C. PCA dimensionality reduction before feature selection.
D. Metadata management with sensitive field tagging. ✓

Justifications:

A. API logging records inference-time requests and responses, not preprocessing decisions about which features were flagged or removed before training.
B. Imputation fills in missing values to improve data quality but does not identify or document sensitive features.
C. PCA reduces dimensionality for modeling efficiency, not for tracking sensitive attribute handling.
D. Metadata management with sensitive field tagging creates a documented record of which features were classified as sensitive and what actions were taken, directly addressing the regulator's request.

Module 10: Continuous Learning and Adaptation

Question M10-Q1

What is the main purpose of continuous learning in AI systems?

A. To ensure models adapt to new data while retaining prior knowledge. ✓
B. To maximize training speed and throughput regardless of accuracy.
C. To reduce model size and parameter count for faster inference at deployment.
D. To eliminate the need for ongoing monitoring, retraining, or manual updates.

Justifications:

A. Continuous learning enables adaptation to new data.
B. Speed is not the priority in continuous learning for AI systems.
C. Not the main purpose of continuous learning.

D. Monitoring is needed even without continuous learning.

Question M10-Q2

Which scenario best illustrates prediction drift?

A. The training dataset becomes corrupted.
B. The outputs of a model shift significantly compared to historical predictions. ✓
C. Input features of a fraud detection system change over time.
D. New features are added to a dataset mid-deployment.

Justifications:

A. Not prediction drift.
B. Prediction drift occurs when model outputs differ over time.
C. That's data drift.
D. Not drift.

Question M10-Q3

What is catastrophic forgetting in continuous learning?

A. When a model loses prior knowledge after retraining on new data. ✓
B. When a system's database becomes corrupted due to storage/hardware failures.
C. When bias testing is skipped during the model validation and evaluation.
D. When model weights cannot be updated due to hardware limitations.

Justifications:

A. Catastrophic forgetting = loss of old knowledge after retraining.
B. Database corruption unrelated.
C. Not bias testing.
D. Not hardware.

Question M10-Q4

Why would a healthcare consortium adopt federated learning instead of centralizing data?

A. To eliminate the need for monitoring and compliance oversight across institutions.
B. To increase training speed by pooling patient data into a single central repository.
C. To reduce compute costs by distributing workloads across organizations.
D. To protect patient privacy while still enabling collective model training. ✓

Justifications:

A. Monitoring is still needed.
B. Pooling reduces privacy.
C. Cost is not the main driver.
D. Federated learning enables training without data sharing.

Question M10-Q5

Which feedback mechanism ensures expert input is weighted more heavily than casual user input?

A. Contextual feedback analysis. ✓
B. Random sampling of feedback.
C. Batch updates every quarter.
D. Consensus by majority voting.

Justifications:

A. Contextual analysis prioritizes expert input.
B. Random ignores expertise.
C. Batching doesn't weight experts.
D. Majority treats all equal.